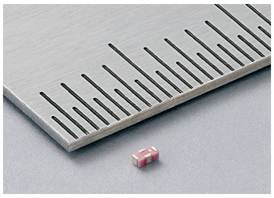
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

World's first parasitic element coupling device for antenna design

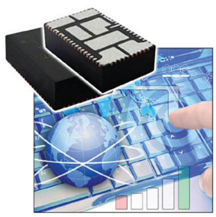
Murata has introduced the world's first parasitic element coupling device that enables better Wi-Fi 6E and Wi-Fi 7 antenna design. It is suitable for designers of smartphones, tablets, network routers, game consoles, and other electronics, enabling them to build efficient antennas for space-constrained devices. The device is created using advanced technology as a surface-mount component measuring 1.0 x 0.5 x 0.35mm. The compact coupling device achieves strong performance without needing magnetic materials. It addresses the issue of reduced coupling in smaller antennas, maintaining the connection between the feeding antenna and parasitic element. By using the device, you can improve antenna matching and reduce performance degradation in wireless communications, even when using long cables.



Murata
https://www.murata.com

Industry's smallest buck regulator module

New Yorker Electronics has launched Vishay's microBRICK synchronous buck regulator module, SiC931, which the company claims to be industry's smallest buck regulator module. The module is suitable for a variety of applications, including POL converters in servers, cloud computing, high-performance computing, desktop computers, industrial automation, motor drives, surveillance systems, consumer electronics, and 5G telecom equipment.



The module is up to 69-percent smaller compared to other solutions. It is highly efficient, with minimal quiescent current consumption, achieving peak efficiency of up to 97 percent. It is particularly effective in reducing energy consumption for FPGAs, ASICs, and SoC core power supplies in data centres, telecom, and industrial applications.

New Yorker Electronics
http://www.newyorkerelectronics.com

High-resolution SWIR image sensor

Sony Semiconductor Solutions Corporation (SSS) has launched IMX992 short-wavelength infrared (SWIR)

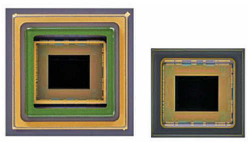
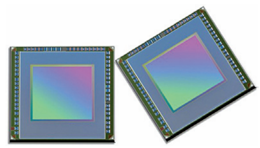


image sensor that has a pixel count of 5.32 effective megapixels. It features a pixel structure for capturing light, enabling imaging across a spectrum ranging from the visible to invisible short-wavelength infrared regions. The applications of the sensor include wafer bonding, defect inspection, and inspecting ingredients and contaminants in food production. The pixel size of 3.45µm is achieved, which is the smallest in its category. The smaller pixel size contributes to a more compact design while still providing a high pixel count. The increased pixel count is crucial for detecting small objects or covering a wide imaging range.

Sony Semiconductor Solutions Corporation
https://www.sony-semicon.com

First iToF sensor with an integrated depth-sensing ISP

Samsung Electronics has introduced ISOCELL Vizion 931, a global shutter sensor. It is the first iToF sensor with an integrated depth-sensing hardware image signal processor (ISP). It can



process images at up to 60 frames per second in QVGA resolution (320 × 240), a resolution used in commercial and industrial settings. The sensor uses backside scattering technology (BST) to enhance light absorption, giving the Vizion 63D high quantum efficiency, reaching 38% at an infrared light wavelength of 940 nanometres (nm). The sensor supports flood and spot lighting modes, extending its measurable distance range from the earlier five metres to 10, enhancing its use in various lighting conditions and distances.

Samsung Electronics
https://www.samsung.com

World's first micrometre-accurate distance radar

OndoSense has launched the first micrometre-accurate distance radar.

The radar sensor is designed for distance measurement, object detection, and positioning, meeting the demands of industrial automation. The OndoSense reach offers millimetre-level